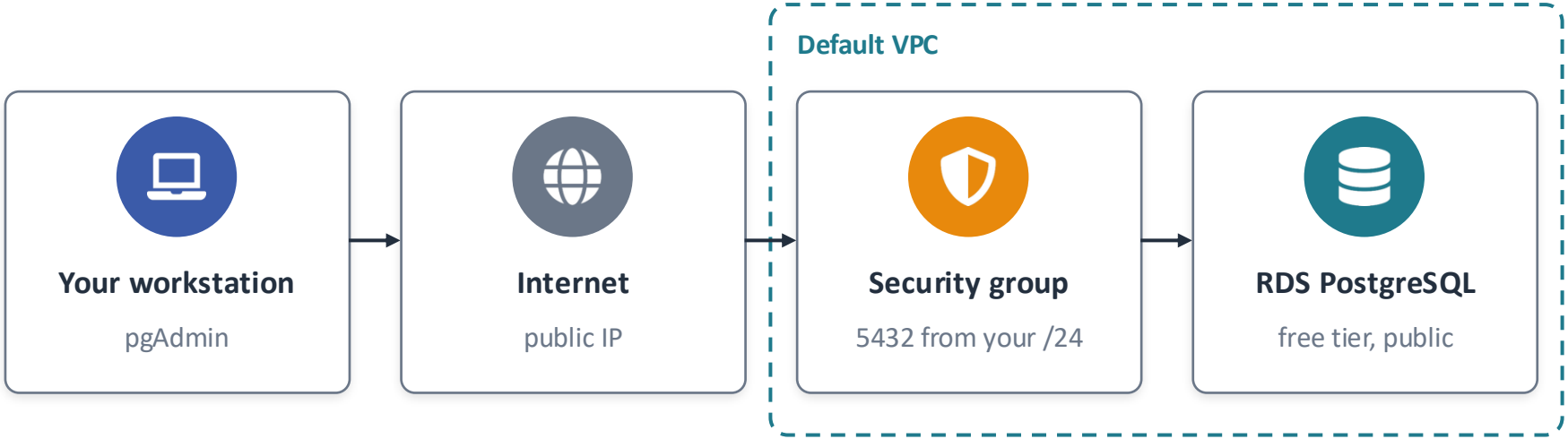
AWS · AMAZON RDS

Create a Free-Tier PostgreSQL Instance on Amazon RDS

Default VPC · Port 5432 restricted to your /24 · Connect with pgAdmin



What we are building



- **Part 1** Create the database in the console
- **Part 2** Open port 5432 to your /24 only
- **Part 3** Connect with pgAdmin
- **Part 4** Clean up so nothing bills

Prerequisites



AWS account

Billing set up. Know whether it is on the credit-based Free plan.



One region

Everyone uses the same one, for example us-east-1.



Default VPC

VPC console, Your VPCs, Default VPC = Yes. Recreate it if deleted.

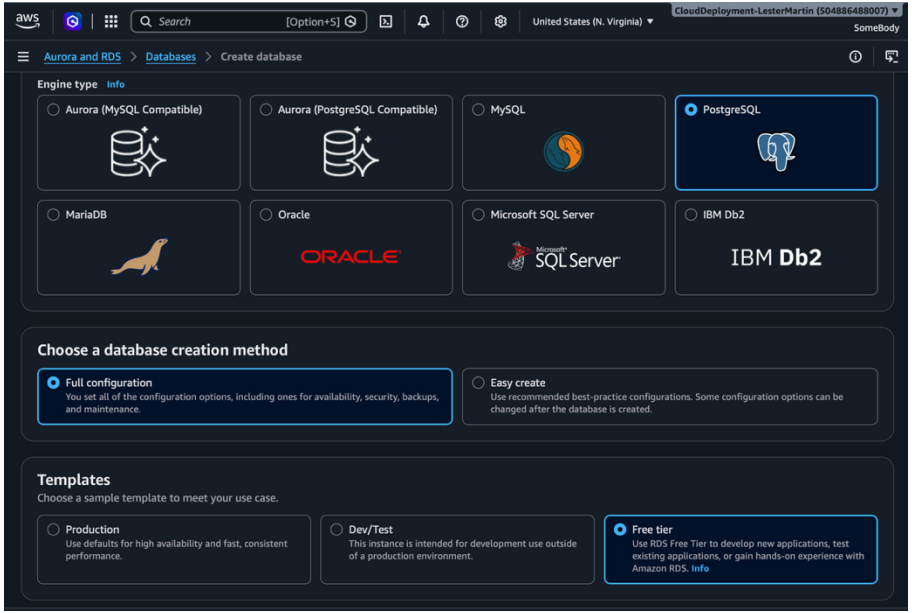


pgAdmin

Already installed on the workstation. Optional: psql.

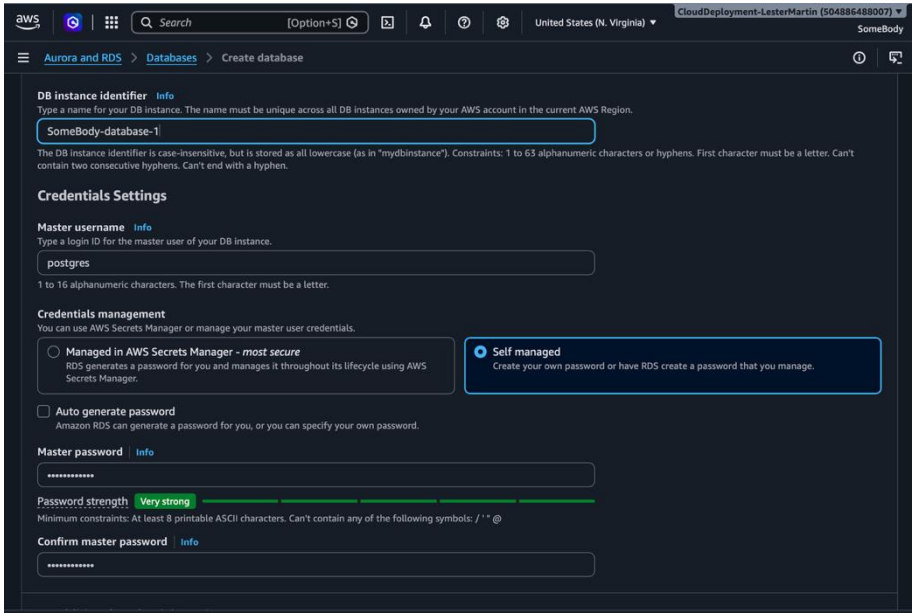
Start the wizard

1. **Sign in** to the console and set your region to **United States (N. Virginia)**.
2. **Open RDS** and choose Databases, then **Create database**.
3. **Full configuration.** Not Express configuration, which hides the networking options.
4. **Engine:** PostgreSQL, not Aurora.
5. **Template:** Free tier. It preselects a free-tier instance and single-AZ.



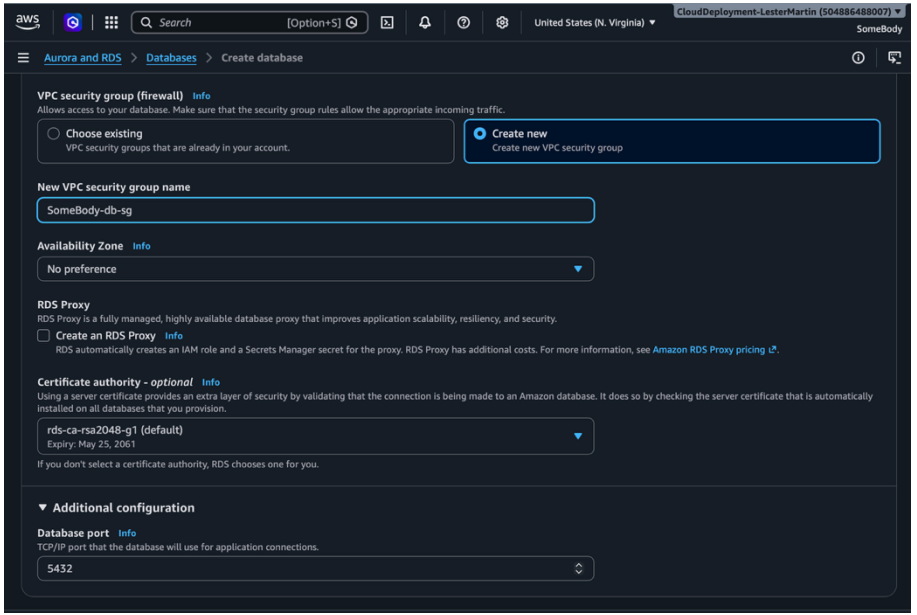
Settings and instance size

6. **Identifier:** *<YourName>*-db-1
7. **Master username:** postgres
8. **Credentials:** Self managed. Secrets Manager is not free.
9. **Password:** strong, entered twice. Write it down.
10. **Additional credentials settings > DB auth options:** Password authentication.
11. **Instance class:** db.t3.micro or db.t4g.micro (free-tier eligible).



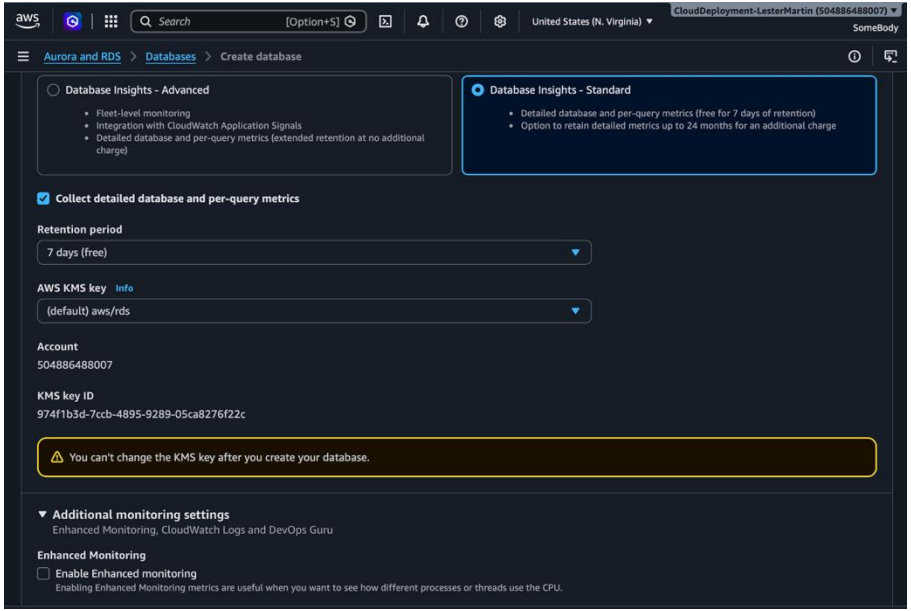
Storage and connectivity

11. **Storage:** General Purpose SSD, 20 GiB (the free-tier maximum).
12. **Additional storage config > Storage autoscaling:** turn it off.
13. **Compute resource:** Don't connect to an EC2 compute resource.
14. **VPC:** the default VPC. Subnet group: default.
15. **Public access:** Yes.
16. Create new, named *<YourName>*-db-sg, **Security group**. Port 5432 under **Additional config**.



Finish and create

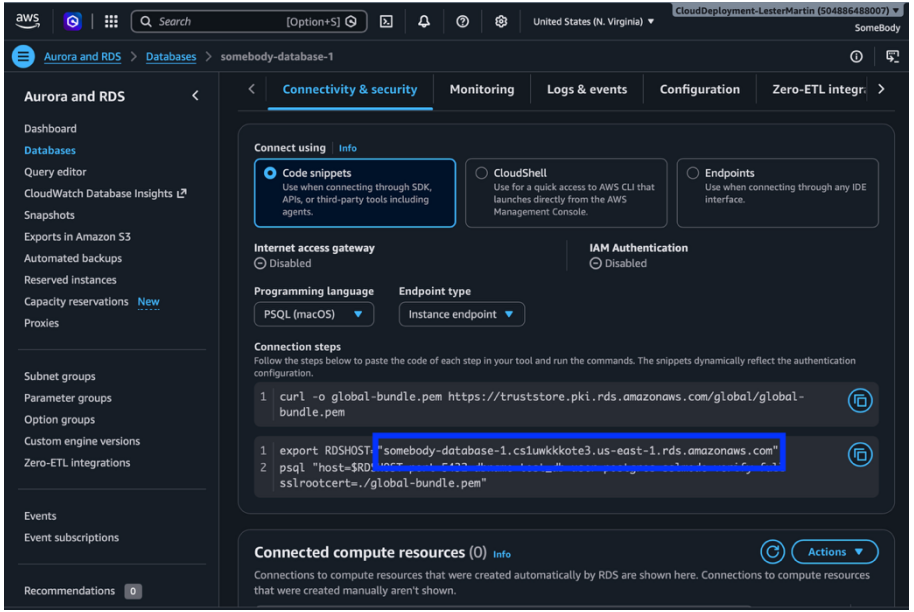
17. **Monitoring:** leave Performance Insights and Enhanced Monitoring off.
18. **Additional configuration:** initial database name test_db.
19. **Backups and protection:** short retention, deletion protection off for easy cleanup.
20. **Check the cost panel,** then click Create database.



Wait, then copy the endpoint

22. **Wait 5 to 10 minutes** until Status = Available.
23. **Open the instance** and find the Connectivity & security tab.
24. **Copy the Endpoint.** You will paste it into pgAdmin in Part 3.

Example endpoint
name-db-1.abc123xyz.us-east-1.rds.amazonaws.com



Step A: find your public IP

Use the public address the internet sees, not your private 192.168.x.x address.

```
curl -s https://checkip.amazonaws.com
```

- Or browse to checkip.amazonaws.com.
- Or in the security group editor, set Source to My IP and copy the address (it fills in a /32).
- Example result: 203.0.113.57

Step B: convert it to a /24

Keep the first three octets, set the last to 0, add /24.

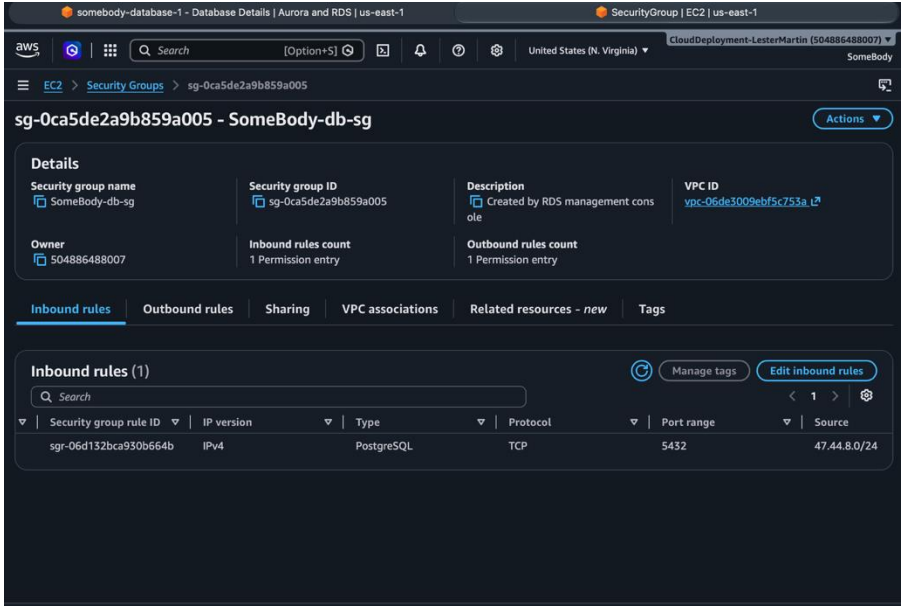
Public IP	CIDR to use	Addresses covered
203.0.113.57	203.0.113.0/24	203.0.113.0 – 203.0.113.255 (256)
198.51.100.214	198.51.100.0/24	198.51.100.0 – 198.51.100.255 (256)

Why 256?
/24 fixes the first 24 bits (3 octets). The last 8 bits are free: $2^8 = 256$ addresses.

```
curl -s https://checkip.amazonaws.com |
  awk -F. '{print $1"."$2"."$3".0/24"}'
```

Step C: add the inbound rule

1. Navigate to the **Security group rules** under the instance's **Connectivity & security** tab.
2. Select ***<YourName>*-db-sg** inbound link to edit inbound rules.
3. **Delete** any wizard-created 5432 rule.
4. **Add rule:** Type PostgreSQL (TCP 5432). Source Custom: your /24, e.g. 203.0.113.0/24.
5. **Description:** Course demo, delete after class. Save rules.



Step D: verify and know the limits

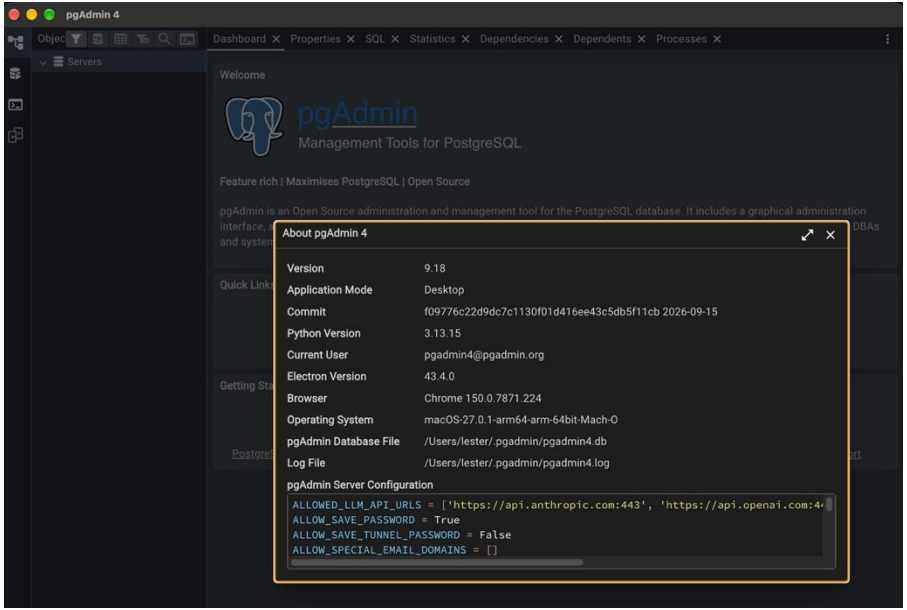
- **Negative test:** try from a phone hotspot. The connection should time out.
- **Your IP can change.** If connections suddenly time out, redo Step A and update the rule.
- **Shared networks:** campus, corporate, or VPN /24s include other people. Run Step A while on the VPN.
- **Not strong security:** the password is the real protection. Production uses private subnets.
- **IPv6:** needs its own rule. Use curl -4 to get an IPv4 address.

Contrast exercise
Try My IP (/32) versus the /24. Discuss the tradeoff.

Bots are fast
Port 5432 open to 0.0.0.0/0 is probed within minutes.

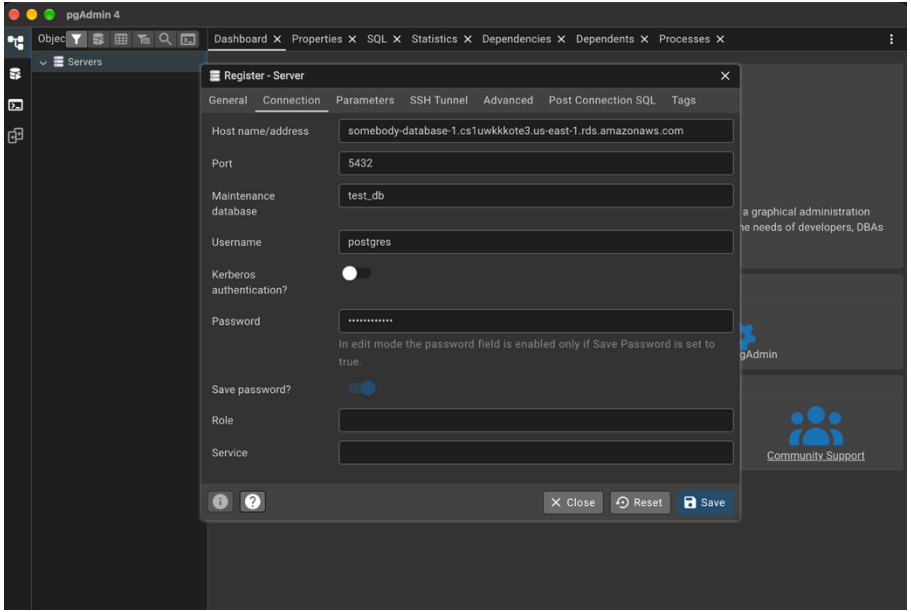
Step A: open your existing pgAdmin

1. **Launch pgAdmin** from the Start menu, Applications folder, or app menu.
2. **Enter the master password** if prompted. It protects saved credentials only.
3. **Show the Object Explorer** (View, Show/Hide Browser Panel) if hidden.
4. **Remove stale entries** from earlier classes under Servers.
5. **Check your version** under Help, About pgAdmin 4. Steps match 8.x and 9.x.



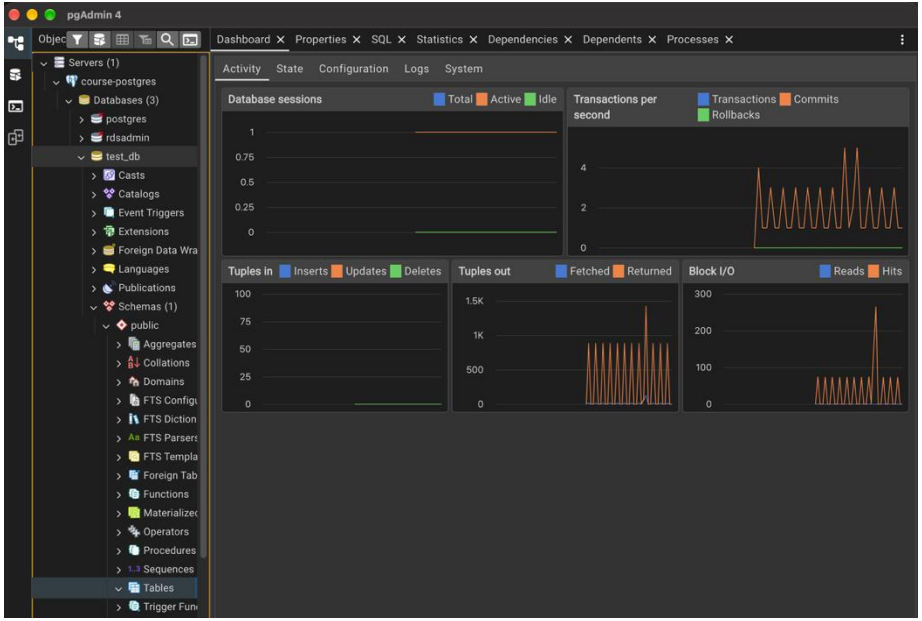
Step B: register the RDS server

1. **Right-click Servers,** then Register, Server.
2. **General tab:** Name = course-postgres (local label only).
3. **Connection tab:** Host = the RDS endpoint. Port = 5432.
4. **Maintenance database:** test_db.
5. **Username / password:** postgres and the master password.



Save and confirm the connection

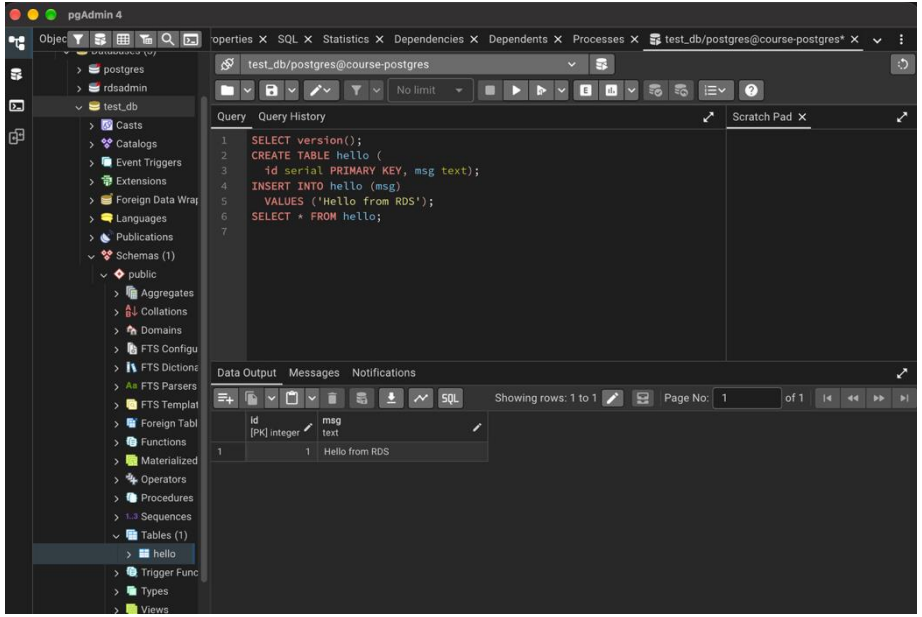
6. **SSL tab (optional):** mode prefer works. Use require to force encryption.
7. **Click Save.** pgAdmin connects immediately.
8. **Expand the tree:** course-postgres, Databases, coursedb, Schemas, public, Tables.
9. **Error instead?** See the troubleshooting slide.



Run your first queries

1. Right-click **test_db,** choose Query Tool (Alt+Shift+Q).
2. **Paste the script,** click Execute (F5).
3. **Results** appear in Data Output; timing in Messages.

```
SELECT version();
CREATE TABLE hello (
  id serial PRIMARY KEY, msg text);
INSERT INTO hello (msg)
  VALUES ('Hello from RDS');
SELECT * FROM hello;
```

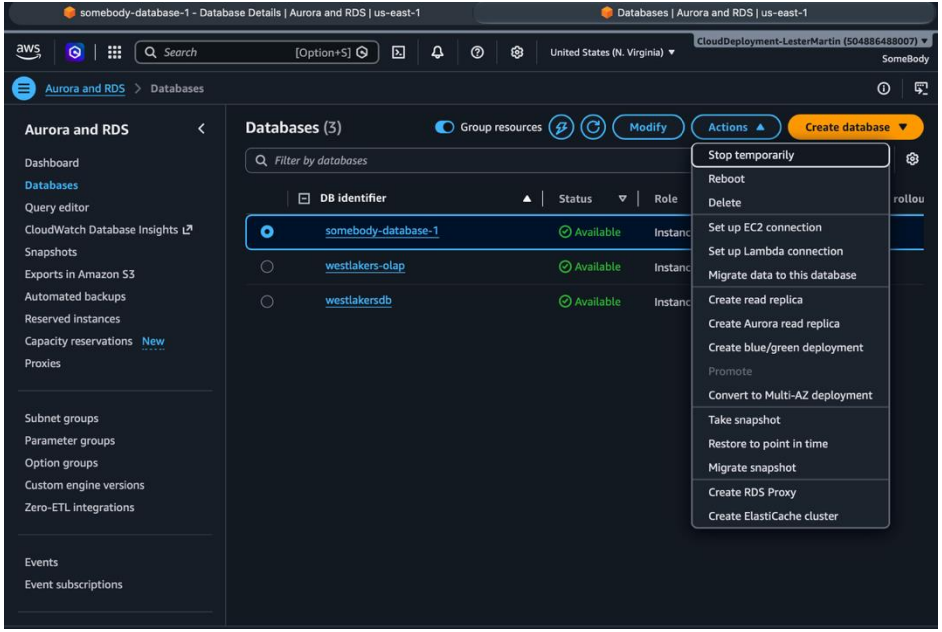


Troubleshooting pgAdmin

Symptom	Likely cause
Connection timed out	Rule missing or your current IP is outside the /24; or Public access is No
Password authentication failed	Wrong password. Reset via RDS, Modify, New master password.
Database coursedb does not exist	Initial name left blank. Set Maintenance database to postgres.
SSL error	Set SSL mode to prefer or require, not disable.
Worked yesterday, fails today	Your public IP changed. Redo Part 2 Step A.
Times out on every rule	Campus or corporate firewall blocks outbound 5432. Try a hotspot.

Delete everything *OR at least STOP instance*

1. **pgAdmin:** right-click the server, Remove Server.
2. **RDS:** select the instance, Actions, Delete.
3. **Uncheck** Create final snapshot and Retain automated backups.
4. **Confirm:** tick the acknowledgment, type delete me.
5. **EC2, Security Groups:** delete course-postgres-sg.
6. **Billing, Free Tier:** confirm no remaining usage.



Key takeaways



Free tier

750 hours a month, 20 GB storage, 20 GB backups. Verify current terms.



Security

A /24 beats 0.0.0.0/0, but the password and private subnets are what protect production.



Cost

Delete the instance. Set a billing alarm or Budget first.